





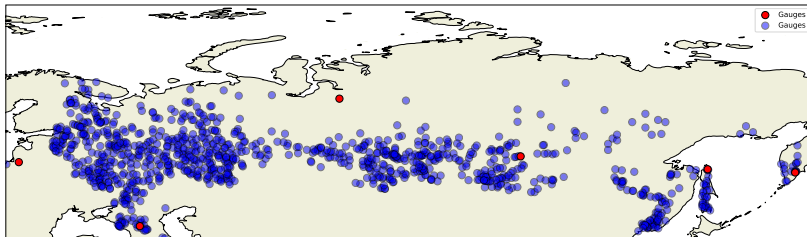
10.1002/hyp.3360060305



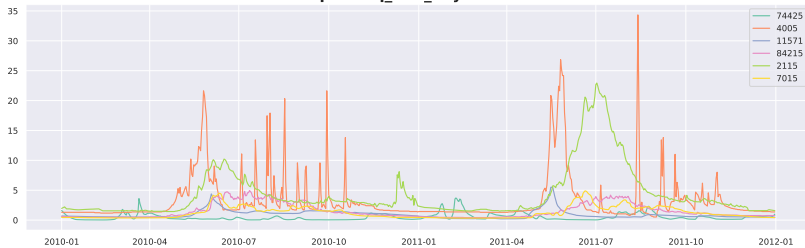
10.31223/X5939N

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)]$$

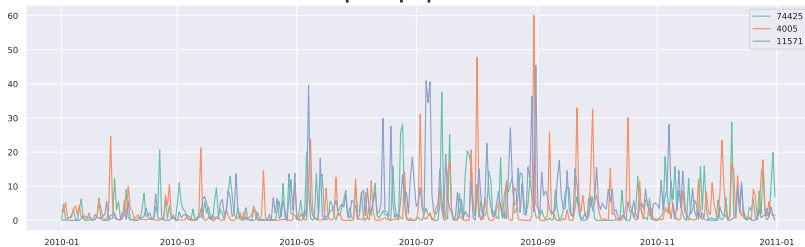
Training River Gauge Locations



Example of q_mm_day time series

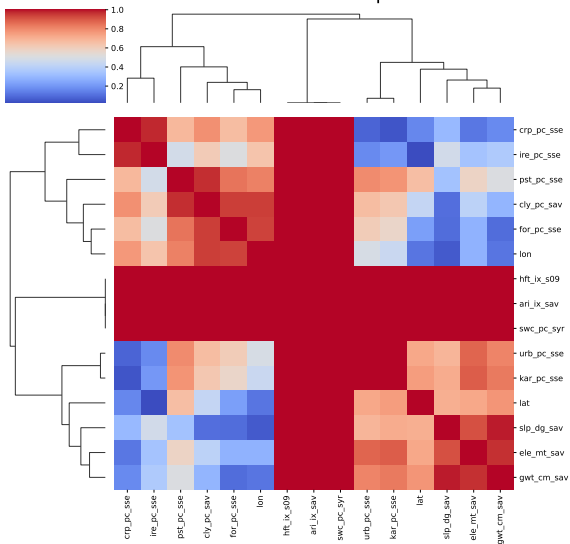


Example of prcp time series



for_pc_sse
crp_pc_sse
pst_pc_sse
ire_pc_sse
urb_pc_sse
kar_pc_sse
ele_mt_sav
slp_dg_sav
ari_ix_sav
gwt_cm_sav
swc_pc_syr
cly_pc_sav
hft_ix_s09
lat
lon

Correlation Heatmap



- ▶ yx
- ▶ $y \in \mathbb{R}^n, x \in \mathbb{R}^{n \times d}$
- ▶ $\mathcal{D} = \{(x_i, y_i) \mid i = 1, \dots, n\}$
- ▶ $\mathcal{F}$
- ▶ $\hat{y}_i = f(x_i), f \in \mathcal{F} \hat{y} = (f(x_1), \dots, f(x_n))$

$$f^* = \underset{f \in \mathcal{F}}{\arg\min} (\mathcal{L}(f, \mathcal{D}) + \Omega(f))$$

$$f(x)=\sum_{l=1}^L\omega_l\mathbb{I}(x\in R_l).$$

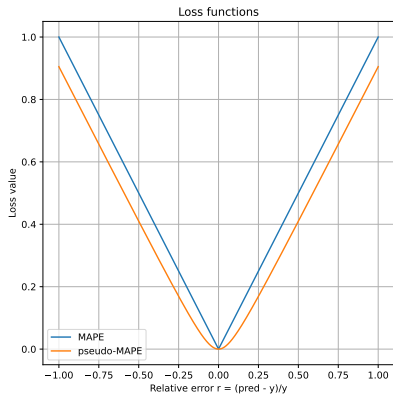
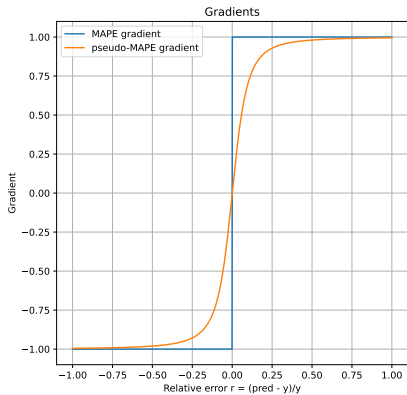
$$\mathcal{I}\mathcal{G}(t,j)=\frac{1}{2}\Big[\frac{(G_L)^2}{H_L+\lambda}+\frac{(G_R)^2}{H_R+\lambda}-\frac{G^2}{H+\lambda}\Big]-\gamma,$$

$$G_{L,R},H_{L,R}$$

$$= \frac{1}{n} \sum_{i=1}^n \left| \frac{\hat{y}_i - y_i}{y_i} \right|$$

$$\delta = \frac{1}{n} \sum_{i=1}^n \left(\sqrt{\left(\frac{\hat{y}_i - y_i}{y_i} \right)^2 + \delta^2} - \delta \right)$$

$$= 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}, \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

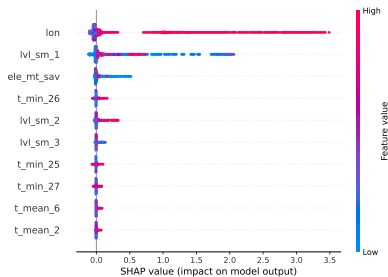
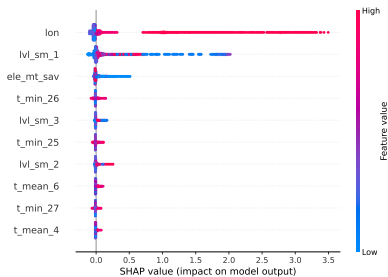
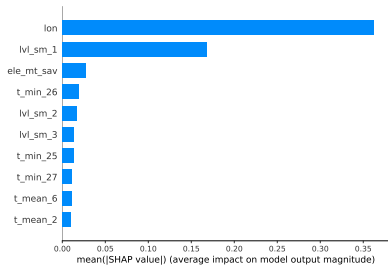
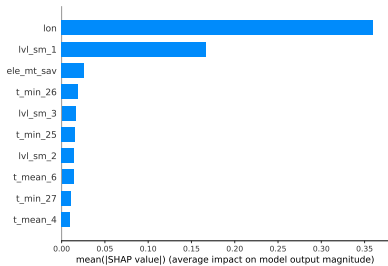


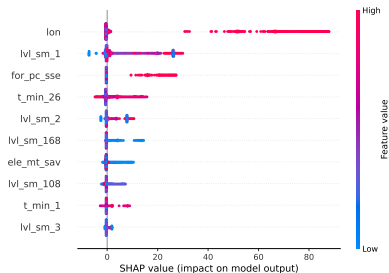
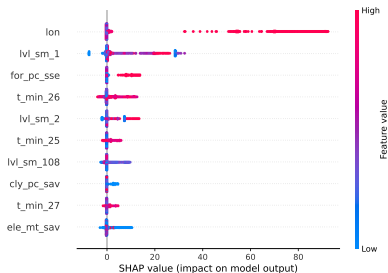
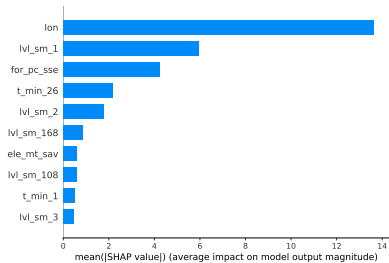
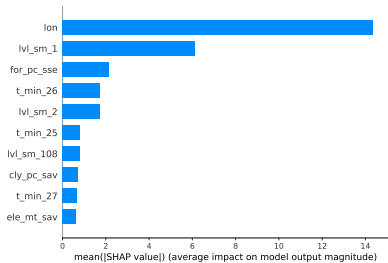
$$f(x) = \frac{1}{B} \sum_{b=1}^B f^{(b)}(x)$$



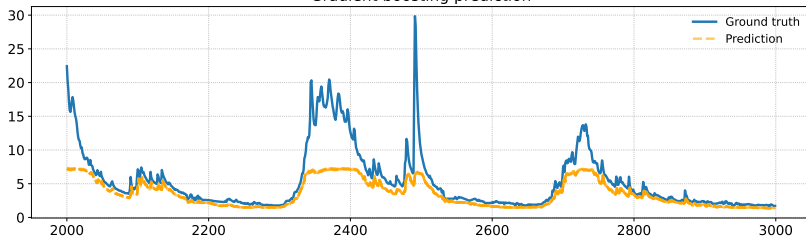
$$f_{(B)}(x) = f^{(0)}(x) + \nu \sum_{b=1}^B f^{(b)}(x)$$







Gradient boosting prediction



Random forest prediction

